Milestone M2 Formal Identifiability & Sensitivity Matrix SVD Report

Project: IIT/SRIC/R/AEH/2026/104 (CMPDI) — *Dynamic Modelling and Digital Twin of Electric Cable Shovel* **Report Status:** Verified against all 12 M2 Simscape .mat trajectories + 4 candidate excitation designs (2026-11-06) **Companion MATLAB Script:** m2_formal_identifiability_svd.m **Primary Sources:** sensitivity_diagnostic.m, M2_sensitivity_diagnostic.mat, M2_04_sweep_wide.mat, M2_04_sweep_narrow.mat, and M2_design_evolution.md (Revisions V7--V10, V24--V25).

1. Mathematical Formulation of the Sensitivity Basis and SVD

1.1 Governing Equations and Exact Partial Derivatives

For the 2-DOF planar dipper--handle subsystem with generalized coordinates q = [q_3, d_4]^T, the nominal rigid-body + friction equations of motion in shovel_params.m and recover_load.m are:

$$\tau_3(t) = \underbrace{(I_{zz} + M_d r^2)\ddot{q}_3}_{\text{Inertial}} + \underbrace{2 M_d r \dot{d}_4 \dot{q}_3}_{\text{Coriolis}} + \underbrace{M_d g r \cos q_3}_{\text{Gravity}} + \underbrace{f_{v3}\dot{q}_3 + f_{c3}\tanh(\dot{q}_3/\epsilon_3)}_{\text{Friction}} - \underbrace{s_{\mathrm{bail}}(F_y\cos q_3 - F_x\sin q_3)}_{Q_{q_3,\mathrm{ext}}}, f_4(t) = \underbrace{M_d \ddot{d}_4}_{\text{Inertial}} - \underbrace{M_d r \dot{q}_3^2}_{\text{Centrifugal}} + \underbrace{M_d g \sin q_3}_{\text{Gravity}} + \underbrace{f_{v4}\dot{d}_4 + f_{c4}\tanh(\dot{d}_4/\epsilon_4)}_{\text{Friction}} - \underbrace{(F_x\cos q_3 + F_y\sin q_3)}_{Q_{d_4,\mathrm{ext}}},$$

where $r(t) = d_4(t) + c = d_4(t) - 1.32\ \mathrm{m}$ is the dipper--handle composite center-of-gravity (COG) lever arm and $s_{\mathrm{bail}}(t) = d_4(t) + 1.50\ \mathrm{m}$ is the hoist bail pin radius from the saddle pivot.

Differentiating the saddle residual with respect to the external vertical load amplitude F_y and each physical model parameter $\theta = [M_d, I_{zz}, c, f_{v3}, f_{c3}]^T$ yields the local sensitivity basis functions evaluated along any observed trajectory $(q(t), \dot{q}(t), \ddot{q}(t))$:

1. **External Vertical Load Sensitivity on τ_3** ($\mathrm{N\cdot m / N} = \mathrm{m}$): $\phi_{F_y,3}(t) = \frac{\partial Q_{q_3,\mathrm{ext}}}{\partial F_y} = (d_4(t) + 1.50)\cos q_3(t)$.
2. **Full Dipper Mass Sensitivity on τ_3 (Revision V24)** ($\mathrm{N\cdot m / kg} = \mathrm{m^2/s^2}$): $\phi_{M_d,3}(t) = \frac{\partial \tau_3}{\partial M_d} = \underbrace{r(t)^2\ddot{q}_3(t)}_{\phi_{M_d,\mathrm{inert},3}} + \underbrace{2r(t)\dot{d}_4(t)\dot{q}_3(t)}_{\phi_{M_d,\mathrm{Cor},3}} + \underbrace{g r(t)\cos q_3(t)}_{\phi_{M_d,\mathrm{grav},3}}$.
3. **Rotational Inertia Sensitivity on τ_3** ($\mathrm{N\cdot m / (kg\cdot m^2)} = \mathrm{rad/s^2}$): $\phi_{I_{zz},3}(t) = \frac{\partial \tau_3}{\partial I_{zz}} = \ddot{q}_3(t)$.
4. **COG Offset Sensitivity on τ_3** ($\mathrm{N\cdot m / m} = \mathrm{N}$, holding M_d

fixed): $\phi_{c,3}(t) = \frac{\partial \tau_3}{\partial c} = M_d\left(2r(t)\ddot{q}_3(t) + 2\dot{d}_4(t)\dot{q}_3(t) + g\cos q_3(t)\right)$.

5. **Viscous & Coulomb Friction Sensitivities on τ_3**: $\phi_{f_{v3},3}(t) = \dot{q}_3(t) \quad [\mathrm{rad/s}]$, $\qquad \phi_{f_{c3},3}(t) = \tanh(\dot{q}_3(t)/\epsilon_3) \quad [-]$.
6. **Crowd Channel (d_4) Sensitivities**: $\phi_{F_y,4}(t) = \sin q_3(t) \quad [\mathrm{N/N}]$, $\qquad \phi_{M_d,4}(t) = \ddot{d}_4(t) - r(t)\dot{q}_3(t)^2 + g\sin q_3(t) \quad [\mathrm{N/kg}]$.

1.2 Why Unscaled SVD Misleads and How Column Scaling Must Be Formulated

Because the columns of the raw sensitivity matrix $S_{\mathrm{raw}} = [\phi_{F_y}, \phi_{M_d}, \phi_{I_{zz}}, \phi_{f_{v3}}] \in \mathbb{R}^{N\times p}$ carry incompatible physical units (m, $\mathrm{m^2/s^2}$, $\mathrm{rad/s^2}$, $\mathrm{rad/s}$), the raw singular values $\sigma_i(S_{\mathrm{raw}})$ change arbitrarily under unit conversions (e.g., kN vs. N or t vs. kg). Moreover, because the linear recovery regression in recover_load.m includes a free intercept, constant offsets are absorbed by the intercept and must be projected out via the centering operator $P_{\perp \mathbf{1}} = I_N - \frac{1}{N}\mathbf{1}\mathbf{1}^T$:
$S_0 = P_{\perp \mathbf{1}} S_{\mathrm{raw}}$.
We therefore evaluate two rigorous, scale-consistent SVD formulations:

1. **Unit-Column-Norm (Geometric Collinearity) SVD**: $\tilde{S} = S_0 D_{\mathrm{norm}}^{-1}, \qquad D_{\mathrm{norm}} = \mathrm{diag}(\Vert{}S_{0,\bullet 1}\Vert{}_2, \dots, \Vert{}S_{0,\bullet p}\Vert{}_2)$. Here $\tilde{S}^T \tilde{S} = R_{\mathrm{corr}}$ is the exact mean-removed Pearson correlation matrix, the singular values satisfy $\sum_{i=1}^p \sigma_i(\tilde{S})^2 = p$, the condition number $\kappa_2(\tilde{S}) = \sigma_{\max}(\tilde{S})/\sigma_{\min}(\tilde{S}) \ge 1$ is strictly dimensionless and unit-invariant, and the **principal subspace angle** $\theta_{\min}$ between the external-load column $\tilde{s}_1$ and the model-error subspace $\mathcal{M} = \mathrm{span}(\tilde{s}_2, \dots, \tilde{s}_p)$ is: $\sin\theta_{\min} = \Vert{}(I_N - P_{\mathcal{M}})\tilde{s}_1\Vert{}_2 = \sqrt{1 - R_{\mathrm{confound}}^2}, \qquad \mathrm{VIF}_1 = (R_{\mathrm{corr}}^{-1})_{11} = \frac{1}{\sin^2\theta_{\min}}$.
2. **Characteristic Physical Perturbation SVD**: $S_{\mathrm{phys}} = \frac{1}{\sqrt{N}} S_0 W_{\mathrm{phys}}, \qquad W_{\mathrm{phys}} = \mathrm{diag}(\Delta F_{\mathrm{nom}}, \Delta M_{d,\mathrm{nom}}, \Delta I_{zz,\mathrm{nom}}, \Delta f_{v3,\mathrm{nom}})$, with $\Delta F_{\mathrm{nom}} = 100\ \mathrm{kN}$, $\Delta M_{d,\mathrm{nom}} = 0.10\, M_d = 5{,}430\ \mathrm{kg}$, $\Delta I_{zz,\mathrm{nom}} = 0.10\, I_{zz} = 28{,}790\ \mathrm{kg\cdot m^2}$, and $\Delta f_{v3,\mathrm{nom}} = 0.10\, f_{v3} = 380\ \mathrm{N\cdot m\cdot s/rad}$. Every entry of S_{phys} has units of $\mathrm{N\cdot m}$ (RMS torque), so $\sigma_i(S_{\mathrm{phys}})$ directly measures the RMS saddle torque (in $\mathrm{N\cdot m}$) produced by each orthogonal principal direction!

2. Comprehensive SVD & Subspace Angle Results Across All 16 Trajectories

The table below reports the exact verified metrics across all 12 M2 Simscape .mat archives (N = 2{,}501 post-trim samples, t \in [5.0, 30.0]\ \mathrm{s}) and 4 analytical candidate trajectories designed to test frequency decoupling and short-extension operation:

Trajectory Archive / Excitation Case	\Delta d_4 Span (\mathrm{m})	\Delta\rho_{\mathrm{station}} Spread	Gravity : Dynamic Ratio	r(\text{Load}, M_{d,\mathrm{grav}})	r(\text{Load}, M_{d,\mathrm{full}})	2-Col Angle \theta_{\min} (Load vs. M_d)	2-Col Unit \kappa_2(\tilde{S})	4-Col Unit \kappa_2(\tilde{S})	Joint 2-Ch (\tau_3+f_4) Angle \theta_{\min}	Joint 2-Ch Unit \kappa_2
Wide_w0.314 (P02/P04)	3.9990	0.2202	40.25:1	0.538886	0.822216	34.6928^\circ	3.20	171.62	10.64^\circ	10.74
Wide_w0.500 (P04)	3.9975	0.2201	16.59:1	0.475807	0.931905	21.2662^\circ	5.33	104.62	10.02^\circ	11.41
Wide_w0.700 (P04)	3.9951	0.2200	8.31:1	0.469873	0.981138	11.1460^\circ	10.25	83.67	32.82^\circ	3.40
Wide_w1.000 (P04)	3.9900	0.2197	4.16:1	0.479330	0.996529	4.7752^\circ	23.98	71.35	79.89^\circ	1.19
Wide_w1.360 (P04)	3.9815	0.2191	2.20:1	0.494885	0.999231	2.2464^\circ	51.00	68.11	76.92^\circ	1.26
Narrow_Bl_w0.314 (P04b/P06)	0.5399	0.0213	42.71:1	0.999973	0.999992	0.2302^\circ	497.71	3794.23	5.10^\circ	22.45
Narrow_Bl_w0.500 (P04b)	0.5397	0.0213	17.57:1	0.999973	1.000000	0.0543^\circ	2110.04	2584.33	1.65^\circ	69.52
Narrow_Bl_w0.700 (P04b)	0.5393	0.0213	8.84:1	0.999974	0.999990	0.2513^\circ	455.98	1949.91	8.36^\circ	13.68
Narrow_Bl_w1.000 (P04b)	0.5387	0.0213	4.40:1	0.999973	0.999956	0.5396^\circ	212.37	1445.36	19.51^\circ	5.82
Narrow_Bl_w1.360 (P04b)	0.5375	0.0212	2.33:1	0.999974	0.999915	0.7455^\circ	153.71	1213.25	29.70^\circ	3.77

Trajectory Archive / Excitation Case	\Delta d_4 Span (\mathrm{m})	\Delta\rho_{\mathrm{station}} Spread	Gravity : Dynamic Ratio	r(\text{Load}, M_{d,\mathrm{grav}})	r(\text{Load}, M_{d,\mathrm{full}})	2-Col Angle \theta_{\min} (Load vs. M_d)	2-Col Unit \kappa_2(\tilde{S})	4-Col Unit \kappa_2(\tilde{S})	Joint 2-Ch (\tau_3+f_4) Angle \theta_{\min}	Joint 2-Ch Unit \kappa_2
Narrow_Centered_8p75 (P06)	0.5399	0.0276	48.48:1	0.999955	0.999976	0.3958^\circ	289.53	3312.33	6.55^\circ	17.47
Medium_Stroke_8p75_pm1m (00_design)	1.9995	0.1040	45.58:1	0.996677	0.996654	4.6884^\circ	24.43	796.17	7.61^\circ	15.03
Decoupled_Wide (wq3=0.314, wd4=0.71, d4=8.75+-2.0)	4.0000	0.2203	31.86:1	0.989096	0.994031	6.2631^\circ	18.28	36.15	21.70^\circ	5.22
Short_Extension_Admissible (wq3=0.314, wd4=0.71, d4=7.25+-1.5)	3.0000	0.2570	41.14:1	0.984628	0.990611	7.8575^\circ	14.56	47.11	19.80^\circ	5.73
Short_Extension_InPhase (wq3=0.314, wd4=0.314, d4=7.25+-1.5)	3.0000	0.2570	51.61:1	0.663780	0.828033	34.1028^\circ	3.26	246.53	12.12^\circ	9.42

Trajectory Archive / Excitation Case	\Delta d_4 Span (\mathrm{m})	\Delta\rho_{\mathrm{station}} Spread	Gravity : Dynamic Ratio	r(\text{Load}, M_{d,\mathrm{grav}})	r(\text{Load}, M_{d,\mathrm{full}})	2-Col Angle \theta_{\min} (Load vs. M_d)	2-Col Unit \kappa_2(\tilde{S})	4-Col Unit \kappa_2(\tilde{S})	Joint 2-Ch (\tau_3+f_4) Angle \theta_{\min}	Joint 2-Ch Unit \kappa_2
Narrow_Bl_Decoupled (wq3=0.314, wd4=0.95, d4=9.77+-0.27)	0.5400	0.0213	42.88:1	0.999563	0.999740	1.3059^\circ	87.74	142.40	7.36^\circ	15.55

3. Physical Interpretation of the Four Key Identifiability Findings

1. **Why Narrow-Stroke Operation Collapses Single-Channel (\tau_3) Identifiability**:
   - On the wide baseline trajectory (M2_04_w0.314.mat, $d_4 \in [6.7505, 10.7495]\ \mathrm{m}$, $\Delta d_4 = 3.9990\ \mathrm{m}$), the station ratio $\rho_{\mathrm{station}}(d_4) = (d_4 + 1.50)/(d_4 - 1.32)$ varies from 1.2991 to 1.5192 ($\Delta\rho = 0.2201$), yielding a 34.6928° angle between $\phi_{F_y,3}$ and $\phi_{M_d,\mathrm{full},3}$ and a well-conditioned 2-column matrix $\kappa_2(\tilde{S}) = 3.20$.
   - On the Bi et al. operational-band trajectory (M2_04n_w0.314.mat, $d_4 \in [9.5001, 10.0399]\ \mathrm{m}$, $\Delta d_4 = 0.5399\ \mathrm{m}$), $\rho_{\mathrm{station}}(d_4)$ varies by only $\Delta\rho = 0.0221$ (1.3234 to 1.3455). Consequently, the angle between external load and full dipper mass error collapses $150.7\times$ to $\theta_{\min} = 0.2302^\circ$ ($r = 0.99999193$), and the 2-column condition number explodes from 3.20 to 497.71 ($\mathrm{VIF} = 61{,}927$)!
2. **Why Coupled In-Phase Frequency Sweeping (P04 / P04b) Worsens Rather Than Improves Wide-Stroke Conditioning**:
   - In P04 (M2_04_w0.314.mat through M2_04_w1.360.mat), increasing ω_{traj} from 0.314 to $1.360\ \mathrm{rad/s}$ successfully boosts the dynamic-to-gravity mass sensitivity ratio $18.27\times$ (from 2.48% to 45.38%, reducing the gravity-to-dynamic ratio from 40.25:1 to 2.20:1).
   - However, because ω_{traj} drives both $q_3(t)$ and $d_4(t)$ at the same frequency while $F_y(t) = F_0\sin(\omega_F t)$ remains fixed at $\omega_F = 2\pi/9 = 0.6981\ \mathrm{rad/s}$, the modulation product $r(t)\cos q_3(t)$ and inertial acceleration $r(t)^2\ddot{q}_3(t)$ become increasingly collinear with $s_{\mathrm{bail}}(t)\cos q_3(t)$ as ω_{traj} increases! Thus $\theta_{\min}$ on the wide stroke shrinks from 34.6928° ($\kappa_2 = 3.20$) at $\omega = 0.314\ \mathrm{rad/s}$ to 2.2464° ($\kappa_2 = 51.00$) at $\omega = 1.360\ \mathrm{rad/s}$.

3. **Why Adding \phi_{I_{zz}} and \phi_{c} Creates Exact Multicollinearity on Sinusoidal Trajectories**:
    - Expanding \phi_{c,3}(t) = M_d(2r\ddot{q}_3 + 2\dot{d}_4\dot{q}_3 + g\cos q_3) shows that at nearly constant r(t) \approx \bar{r}, \phi_{c,3}(t) \approx \frac{M_d}{\bar{r}}\phi_{M_d,3}(t) + M_d\bar{r}\phi_{I_{zz},3}(t). Including \phi_c alongside (\phi_{F_y}, \phi_{M_d}, \phi_{I_{zz}}) raises the 5-column condition number to \kappa_2(\tilde{S}) \in [667, 1.49\times 10^5]. Thus (M_d, I_{zz}, c) cannot be identified independently from single-channel saddle torque without changing the mechanical configuration or using the M1 base-parameter grouping \theta = [M_d,\ M_d c,\ I_{zz} + M_d c^2]^T.
4. **How Joint Two-Channel (\tau_3 + f_4) Estimation Breaks the Load-vs-Mass Degeneracy**:
    - On the saddle channel \tau_3, vertical bail load acts through lever arm s_{\mathrm{bail}} = d_4 + 1.50\ \mathrm{m} while gravity mass error acts through r_{\mathrm{COG}} = d_4 - 1.32\ \mathrm{m} (ratio \rho_{\mathrm{station}} \approx 1.334 on the BI band).
    - On the crowd channel f_4, vertical bail load contributes F_y\sin q_3 while dipper mass error contributes \Delta M_d(\ddot{d}_4 - r\dot{q}_3^2 + g\sin q_3) (ratio 1/g in the quasi-static limit, with **zero** lever-arm factor!).
    - Stacking both channels (\tau_3, \bar{s} f_4) into a joint 2-channel sensitivity matrix increases the subspace angle on the narrow BI band from 0.2302^\circ to 5.10^\circ at \omega = 0.314\ \mathrm{rad/s} (\kappa_2 drops 22.2\times from 497.71 to 22.45) and to **29.70^\circ** at \omega = 1.360\ \mathrm{rad/s} (\kappa_2 = 3.77)! On the wide stroke at \omega = 1.000\ \mathrm{rad/s}, joint 2-channel conditioning reaches \kappa_2 = 1.19 (\theta_{\min} = 79.89^\circ).